ACTL 3141 ASSIGNMENT T1 2025

DANIEL DEVARIS

z5480856

April 2025

1. Tasks

1.1. Descriptive analysis of hospitalised patient profiles. To understand the data, I gathered basic summary statistics to develop a general understanding of the data. I first wanted to understand the relationship with age. I noticed that the mean age of hospitalisation for Covid patients was 68.76, whereas, the mean age for Influenza patients was 42.60. This stark contrast prompted me to do some further investigation. To do so, I constructed the following histograms of age to better understand the distribution of patient age.

Figure 1. Influenza Patient Age.

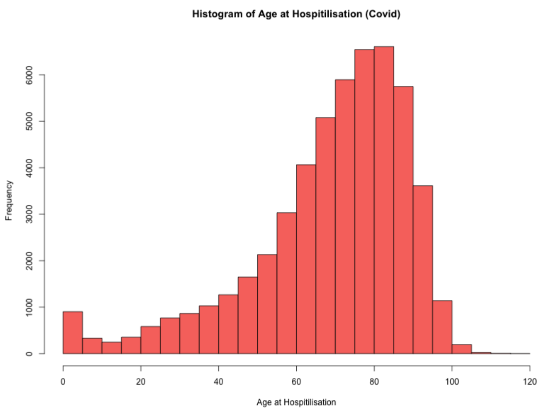

Figure 2. Covid Patient Age.

We can see in Figure 1 that the distribution of Influenza patient age is symmetrical when excluding the large spike in child patients, whereas, in Figure 2, Covid patient age seems to be more negatively skewed, with much less child patients suffering from Covid. We also see that the Covid data is slightly negatively skewed indicating that it effects older members of the population to a greater extent.

Wanting to investigate further, I created the following histograms that separate the data into two groups, whether or not they passed from their illness while in hospital. Firstly, I found the mean of Influenza patients to be 56.7 vs 39.25 when passing or surviving, compared to mean Covid patient age being 74.2 vs 65.7 when passing or surviving. By inspecting Figure 3, we can see that although there is a high level of child Influenza, many of them survive, with the distribution of patients age when passing from Influenza is also slightly negatively skewed. We can also see in Figure 4 that the difference in distribution between those who passed or survived from Covid is relatively similar, with a slightly larger negative skew for patients who passed.

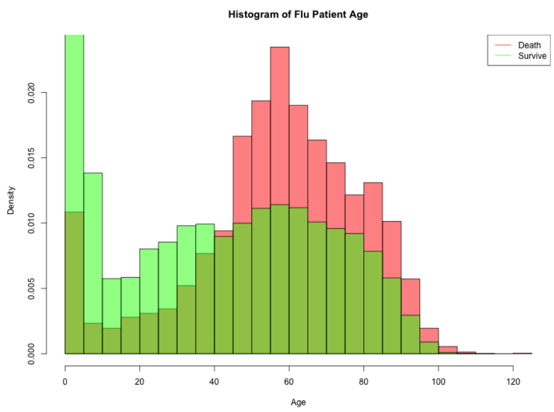

Figure 3. Influenza Patient Age Comparison.

Figure 4. Covid Patient Age Comparison.

I was also intrigued by the distribution of patients with respect to their time spent in hospital and whether or not they passed away. Figures 5 and 6 both demonstrate that both Influenza and Covid patients time in hospital is negatively exponential and does not differ between patients that passed or not.

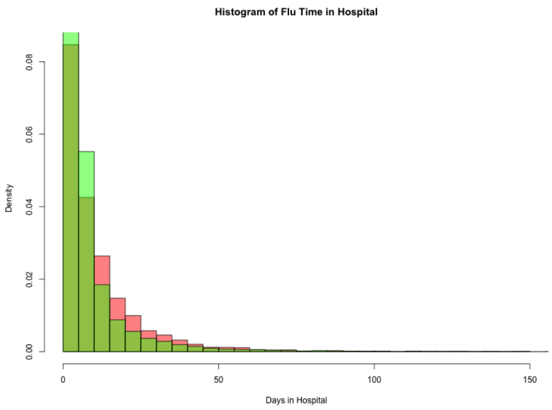

FIGURE 5. Influenza Patient Time in Hospital Comparison.

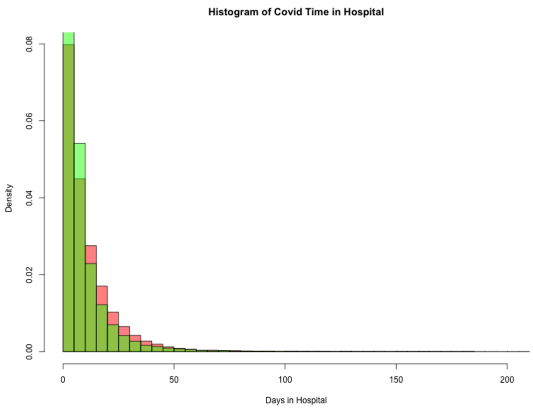

FIGURE 6. Covid Patient Time in Hospital Comparison.

Next, I was intrigued by the ways various risk factors could have an influence on the death of patient. Firstly, I wanted to investigate the proportion of patients with or without a certain condition, as seen in Table 1. Notably, we can see the much larger proportion of death outcomes for Covid patients compared to Influenza patients. We can also see a higher proportion of cardiovascular diseases in Covid patients to Influenza patients and a higher proportion of Covid patients with the Covid vaccine compared to Influenza patients with the Influenza vaccine. These findings may indicate the key differences in severity of the two conditions, which will prove influential for insurance pricing mechanisms.

X_i	Covid	Influenza
Death	0.364	0.193
Cardio	0.520	0.216
Pneumo.	0.0931	0.151
Immuno	0.0736	0.0556
Renal	0.0869	0.0388
Obesity	0.0790	0.0614
Vaccinated	0.781	0.205

TABLE 1. Proportion of each Group with X_i.

1.2. **Survival analysis of Influenza patients.** Next, we will conduct survival analysis techniques on the Influenza patients. I decided to fit a KM survival fit to estimate the survival function of influenza patients over their time in hospital as it is reasonable to assume that for the vast majority lives are independent and that the censoring is non-informative. However, we must note that family members in hospital together may be correlated or individuals that chose to leave hospital if they want a comfortable death but this would not be the norm, thus, KM estimates are still useful. We can also estimate survival functions based on predictor variables. The predictor variables that I am interested in were race, age, obesity, cardiovascular disease and vaccination status. I decided to focus on these variables as they seem to be the most notable; race and age being traditional variables to investigate and obesity and vaccination status as some of the variables with a high proportion in this population.

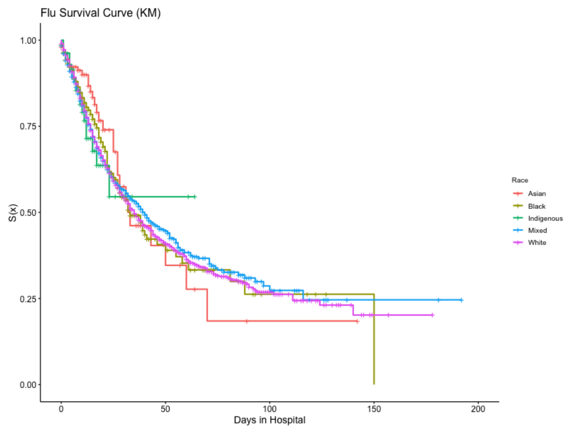

FIGURE 7. KM Survival Estimate (Race).

We can see in Figure 7 that there is relative similarity between all racial groups except for Asians which seem to have a lower survival from 50 days in hospital onward. We can also notice that the Indigenous group has a much shorter curve. With both the Asian and Indigenous population having a much lower number of individuals, 213 and 107 respectively out of 24,439, this deviation in the survival curves could be explained by a lack of data points.

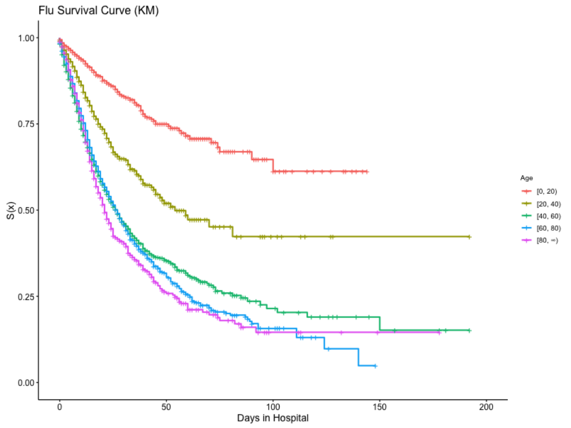

FIGURE 8. KM Survival Estimate (Age).

Now, inspecting Figure 8, we can easily see that older individuals have a higher chance of death, which is to be expected. It is worth noting the spacing between the lines, with ages 0 - 20 having much higher survival than the rest, with ages 40+ all having similar survival curves, but still in the expected order.

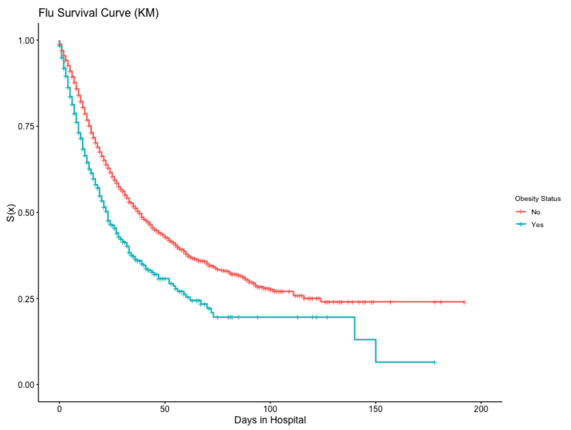

FIGURE 9. KM Survival Estimate (Obesity).

Figure 9 demonstrates the lower chance of surviving influenza when obese, which is to be expected as we can expect obese individuals to be less active and would possess below average respiratory ability which would pose major difficulty in recovering from Influenza.

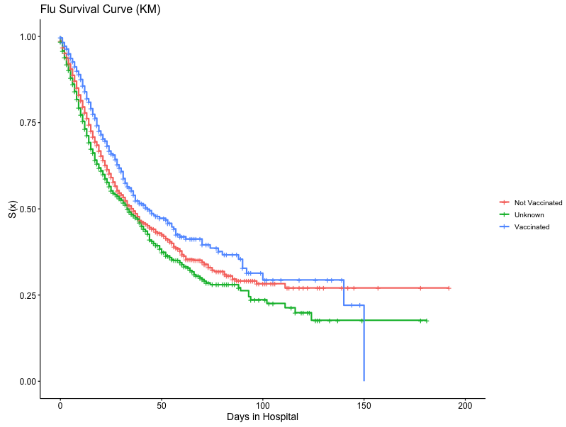

FIGURE 10. KM Survival Estimate (Vaccine).

In the dataset, there were a large number of NAs, especially in the vaccination status of individuals. When dealing with Influenza patients we will only consider the Influenza vaccine. One of the ways I attempted to deal with NAs was to put all individuals less than 6 months old to non vaccinated as individuals less than 6 months old cannot receive the Influenza vaccine. To deal with the rest of the NAs I created a new category `Unknown`. Figure 10 demonstrates that vaccinated individuals have a higher chance of survival than non vaccinated individuals with individuals with an unknown vaccination status having the lowest survival probability of the three.

To further investigate Influenza patient survival, we can fit a Cox regression model to determine the magnitude at which variables effect an individuals survival probability. Using the Cox model as an inference tool rather than a predictive tool, I initially fit a model with all variables. As expected we see that age, obesity and vaccination status to be extremely significant coefficients, impacting survival in the ways that we saw under our KM estimation. Race is also quite significant, however, the other co-morbidities (immuno, cardio, renal pneumo) aren't significant in influencing the survival probability for individuals with Influenza. By then fitting a reduced model including only the significant predictors, the results remain the same with more parsimony.

However, for a Cox model to be a good fit, there are additional assumptions we must satisfy, proportional hazards and the Cox-Snell residual behaviour. To test proportionality we can use `cox.zph()`. As a result of the test we can see that all the variables of significance are in fact proportional demonstrating that the proportional hazard assumption is valid. Further, we can investigate whether the Cox-Snell residuals $E_j \sim Exp(1)$. To do this we can fit a NA estimate of the residuals against the estimated hazard.

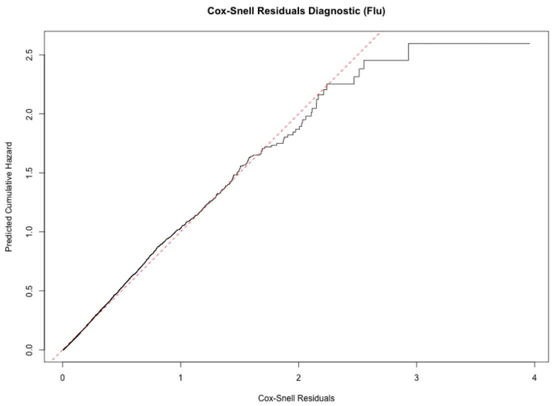

FIGURE 11. Cox-Snell Residual Diagnostic Plot (Influenza).

Figure 11 effectively demonstrates that the residuals roughly follow Exp(1), however, as the residuals increase in size, their estimated cumulative hazard is larger than the true value. We can conclude that a Cox model is a relatively good fit with relatively valid assumptions for this dataset.

From an insurer's perspective when trying to understand how different variables impact an individual's ability to survive Influenza, they should be looking to insure healthy, young and vaccinated individuals, which is to be expected. We can also gather that immuno, cardio, renal and pneumo ailments do not pose great impact to mortality which may help simplify pricing models.

1.3. Evaluating the claim that "COVID-19 is just another flu".

To evaluate the claim that "Covid is just another Flu", we can conduct further survival analysis between the two datasets. Following our analysis on Influenza patients, I also conducted a Cox regression model on Covid patients to try and understand the ways each risk factor influenced survival. The significant coefficients were obesity, age, cardio, immuno, pneumopathy, renal and vaccination status. It is notable to mention how more comorbidities are significant in impacting survival from Covid rather than survival from Influenza. This may indicate that Covid requires much stronger immune systems to recover from than Influenza, indicating a higher chance of survival from Influenza. When inspecting the coefficients, we must note that the coefficients for obesity and cardio are negative, indicating that being obese or having a cardio disease will improve your chance of survival. This result is extremely shocking and unexpected. The rest of the coefficients are expected, with pneumo, renal, immuno and unvaccinated individuals having a significantly lower survival. We can also check that the assumptions of the Cox model hold. In contrast to Influenza, some variables do not have proportional hazards (obesity, pneumo and immuno), which question the validity of using these variables in our model and the overall goodness of fit. We can also plot the Cox-Snell residuals and expect they follow Exp(1). Figure 12 demonstrates that the Cox-Snell residuals for Covid are well behaved, an interesting result following the lack of proportionality. Already, we can see that Covid and Influenza affect patients very differently, with very different results from our Cox regression analysis and the impact of different comorbidities.

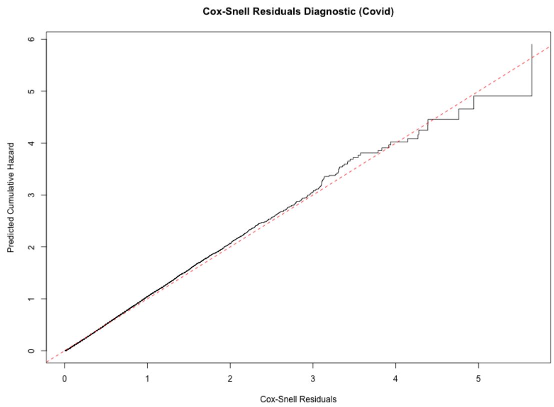

FIGURE 12. Cox-Snell Residual Diagnostic Plot (Covid).

After investigating the risk factors, I wanted to directly compare the mortality of each group. To do this I fit a KM estimate of the survival function to directly compare Covid to Influenza, as seen in Figure 13.

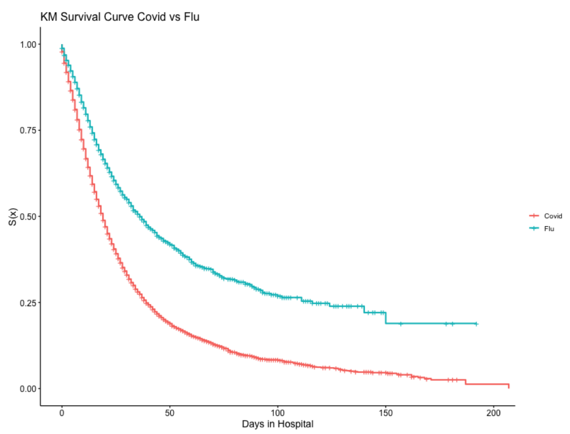

FIGURE 13. KM Estimate Covid vs Influenza.

Figure 13 clearly depicts that the the survival curve for Influenza patients is much higher than Covid patients. To test this more thoroughly, I conducted a log-rank and Peto-Peto test to confirm. Both the log-rank and Peto-Peto tests returned an extremely significant p-value, confirming the claim that mortality is significantly higher for Covid patients than Influenza patients. This claim can be furthered when comparing the estimated baseline hazard functions estimated from our Cox models earlier.

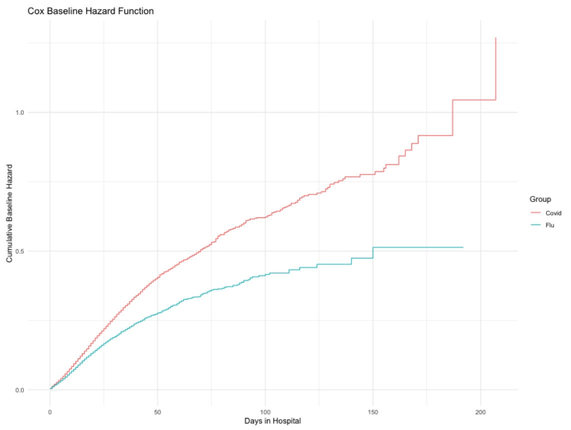

FIGURE 14. Baseline Hazard Covid vs Influenza.

We can see that our Cox models support our KM estimates, indicating that Covid patients have a higher mortality rate than influenza patients, further evidence that Covid is *not* just another flu.
Next, I combined the two datasets and created a Cox model that is only dependent on Covid vs Influenza. The coefficient of Influenza was not surprising, as $\exp \beta = 0.557$, indicating a 45% decrease to the estimated hazard rate when diagnosed with Influenza rather than Covid. It is also not surprising that this coefficient had an extremely significant p-value.

To ensure valid results, we can again test the assumptions of our models. For both KM and Cox, we require independent lives. Once we have combined the two datasets this assumption may be violated as an individual may appear twice in the dataset as an Influenza patient and a Covid patient at different times. This may cause a threat to the validity of our models. Further, we can test the Cox assumptions in similar ways to above. When testing proportionality, `cox.zph()` demonstrates that the hazards between Covid and Influenza are proportional, demonstrating that a Cox model can be valid for this case. Further, we can plot the Cox-Snell residuals against the NA cumulative hazard. Figure 15 depicts this, demonstrating that the Cox-Snell residuals of the combined model when only considering Covid vs Influenza are well behaved, demonstrating a good overall fit of the model.

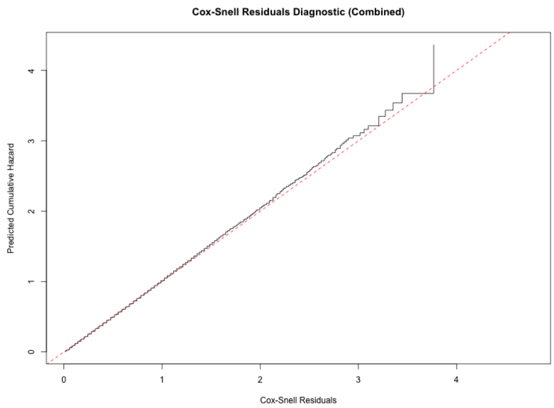

FIGURE 15. Cox-Snell Diagnostic Plot (Combined Dataset).

Ultimately, we can justifiably conclude that Covid is *not* just another Influenza as we can see key differences in the ways risk factors influence mortality rates and the much higher general mortality rates for Covid patients than Influenza patients.

1.4. **Ethical implications of obesity drug use in insurance pricing.** To discuss the ethical implications of using obesity drugs in insurance pricing, we will address two key questions: 1) Should insurers incentivise obesity drug use by offering lower premiums? and 2) What ethical concerns arise from differentiating premiums based on drug use? To answer these questions I will be using the Dobrin (2009) ethical framework.

Step 1 - Identify stakeholders, gather and interpret key facts.
The key stakeholders I am interested in are the policyholders, insurance companies, medical professionals, pharmaceutical companies and society as a whole. It is essential we keep these stakeholders in mind as we assess any ethical implications. Firstly, we can try and assume the interests of each stakeholder. For policyholders and society, we can assume that they are interested in fairness and just decision making on the part of insurance companies. We can assume that the insurance companies are interested in improving the efficiency of their pricing mechanisms whilst maintaining their customers and upholding their reputation as offering fair and non predatory insurance. We can also assume that the pharmaceutical companies are interested in maintaining

profits and constant innovation in biotechnological products. We can also assume that medical professionals have their patients needs as their top priority. It is known that insurance companies use other risk factors age, smoking, and other health conditions to determine life insurance prices. So the question becomes, what type of risk factors are ethically acceptable to use in insurance underwriting.

Step 2 - Identify relevant core values.
Core values that must be accounted for in this discussion are fairness, non-discrimination, autonomy and health equity. We must ask the question, is it fair for individuals to be penalised for their health conditions when many factors beyond their control can influence it, such as genetics and socioeconomic status? We must also consider any form of stigmatisation that could occur as a result of using obesity medication against the obese population. Potentially most importantly, will individuals be free to make their own choices and not risk being financially penalised for them and is obesity medication accessible to all individuals? If these core values are breached then the use of obesity medication can't be used in insurance underwriting.

Step 3 - Courses of Action and Consequences.
Insurance companies can either use obesity medication uptake in their underwriting, or not, as it would be unfair to use it in select cases. We can now analyse the advantages and disadvantages of each course of action. Some advantages for using obesity medication as a risk factor include: more efficient insurance pricing mechanisms and thus financial stability and can incentivise healthy lifestyles. However, some disadvantages include: stigmatisation of obese individuals, penalising groups that may be financial unable to receive the medication and incentivising medication use for lower premiums contradicts our values in step 2 as it is more ethical to incentivise healthy dieting and exercise as a means for weight loss rather than medication use. Advantages of not using obesity medication for insurance can include increased health equity by incentivising healthy lifestyles over medication use and will avoid increasing any stigma around the obese community. Disadvantages of not using obesity medication as a risk factor can include higher premiums for all as a means to offset the additional mortality of obese individuals.

Step 4 - How would this decision be perceived?
If insurers were to use obesity medication as a risk factor, they may receive some public backlash from groups advocating for health equity and anti-discrimination in insurance. Medical professionals may also perceive this decision as positive as it incentivises individuals to improve their health, but may express concerns surrounding the individuals are aiming to improve their health being penalised.

Step 5 - Can you explain your choice to others?
If insurance companies were to use obesity medication as a risk factor, they can explain it through closely linking risk to premium prices, providing a fairer premium associate with an individuals mortality risk. They may also justify through more efficient pricing and greater financial stability for the insurance company. Alternatively, they could justify not using it as a risk factor due to the number of ethical concerns regarding fairness, discrimination, autonomy and health equity that can arise from using it as a risk factor. They may also use it as a way to promote healthy lifestyles to improve health outcomes rather than the use of medication.

Recommendation:
Finally, based on our ethical framework discussion, we will answer the two key questions posed earlier. Firstly, lower premiums should *not* be incentivised through the use of medication as a result of those drugs not being equally accessible and the restriction of individual autonomy due to the financial penalisation of not using obesity medication as an obese individual, further, insurance companies may be inadvertently penalising individuals who aim to lose weight through dieting and exercise simply because they are not using an obesity medication. Secondly, many ethical concerns arise from differentiating premiums on medication use. By differentiating premiums, we are incentivising medication use rather than incentivising better health outcomes. By doing so insurance companies are penalising individuals who cannot afford the related medication, and by increasing their premiums, their financial situation is worsened. Ultimately, the use of obesity medication uptake as a risk factor should not be used in any insurance underwriting due to the vast amount of ethical concerns arising and should instead look to promote healthier lifestyles to reduce premiums as an alternative.

2. References

Dobrin, A., 2009. Business Ethics: The Right Way to Riches. Hindi Granth Karyalay.

3. Appendix

3.1. **Model Outputs.**

```
    > (logrankobe <- survdiff(fluSurv ~ obesity, data = flu, rho = 0))
Call:
survdiff(formula = fluSurv ~ obesity, data = flu, rho = 0)

                  N Observed Expected (O-E)^2/E (O-E)^2/V
obesity=FALSE 22938     4215     4391      7.08       104
obesity=TRUE   1501      505      329     94.62       104

 Chisq= 104  on 1 degrees of freedom, p= <2e-16
> (petopetoobe <- survdiff(fluSurv ~ obesity, data = flu, rho = 1))
Call:
survdiff(formula = fluSurv ~ obesity, data = flu, rho = 1)

                  N Observed Expected (O-E)^2/E (O-E)^2/V
obesity=FALSE 22938     3510     3664      6.44       113
obesity=TRUE   1501      422      268     88.06       113

 Chisq= 113  on 1 degrees of freedom, p= <2e-16
> (logrankrace <- survdiff(fluSurv ~ race, data = flu, rho = 0))
Call:
survdiff(formula = fluSurv ~ race, data = flu, rho = 0)

                    N Observed Expected (O-E)^2/E (O-E)^2/V
race=Asian        213       37     45.4    1.5658    1.6110
race=Black       1158      229    238.4    0.3688    0.3958
race=Indigenous   107       21     20.5    0.0121    0.0124
race=Mixed       6671     1374   1340.8    0.8234    1.1723
race=White      16290     3059   3074.9    0.0824    0.2409

 Chisq= 2.9  on 4 degrees of freedom, p= 0.6
> (petopetorace <- survdiff(fluSurv ~ race, data = flu, rho = 1))
Call:
survdiff(formula = fluSurv ~ race, data = flu, rho = 1)

                    N Observed Expected (O-E)^2/E (O-E)^2/V
race=Asian        213     28.3     37.2    2.1468    2.5856
race=Black       1158    187.8    197.7    0.5014    0.6252
race=Indigenous   107     18.3     17.6    0.0262    0.0304
race=Mixed       6671   1156.9   1111.4    1.8681    3.0796
race=White      16290   2540.4   2567.8    0.2913    0.9925

 Chisq= 5.7  on 4 degrees of freedom, p= 0.2
> (logrankvac <- survdiff(fluSurv ~ influenzaVacine, data = flu, rho = 0))
Call:
survdiff(formula = fluSurv ~ influenzaVacine, data = flu, rho = 0)

                            N Observed Expected (O-E)^2/E (O-E)^2/V
influenzaVacine=No      13760     2622     2609    0.0695     0.158
influenzaVacine=Unknown  5668     1437     1215   40.5482    55.770
influenzaVacine=Yes      5011      661      896   61.8285    77.904
```

```
 Chisq= 105  on 2 degrees of freedom, p= <2e-16
> (petopetovac <- survdiff(fluSurv ~ influenzaVacine, data = flu, rho = 1))
Call:
survdiff(formula = fluSurv ~ influenzaVacine, data = flu, rho = 1)

                            N Observed Expected (O-E)^2/E (O-E)^2/V
influenzaVacine=No      13760     2194     2178     0.105     0.279
influenzaVacine=Unknown  5668     1198      997    40.698    64.920
influenzaVacine=Yes      5011      540      757    61.999    89.877

 Chisq= 121  on 2 degrees of freedom, p= <2e-16
> (logrankage <- survdiff(fluSurv ~ ageBin, data = flu, rho = 0))
Call:
survdiff(formula = fluSurv ~ ageBin, data = flu, rho = 0)

                   N Observed Expected (O-E)^2/E (O-E)^2/V
ageBin=[0, 20)  6789      423     1246     543.8     753.4
ageBin=[20, 40) 4038      459      618      41.0      48.3
ageBin=[40, 60) 5723     1625     1187     161.4     220.3
ageBin=[60, 80) 5412     1469     1143      92.8     124.9
ageBin=[80, )  2477      744      525      91.3     104.8

 Chisq= 951  on 4 degrees of freedom, p= <2e-16
> (petopetoage <- survdiff(fluSurv ~ ageBin, data = flu, rho = 1))
Call:
survdiff(formula = fluSurv ~ ageBin, data = flu, rho = 1)

                   N Observed Expected (O-E)^2/E (O-E)^2/V
ageBin=[0, 20)  6789      359     1046     451.5     723.1
ageBin=[20, 40) 4038      394      528      33.7      45.3
ageBin=[40, 60) 5723     1368      974     159.4     252.4
ageBin=[60, 80) 5412     1200      945      68.5     106.9
ageBin=[80, )  2477      610      438      67.7      89.9

 Chisq= 922  on 4 degrees of freedom, p= <2e-16
```

```
 > summary(coxCovid)
Call:
coxph(formula = covidSurv ~ obesity + ageHosp + cardio + pneumopathy +
    immuno + race + influenzaVacine + covidVacine + renal, data = covid,
    method = "breslow")

  n= 52062, number of events= 18967

                          coef  exp(coef)   se(coef)      z Pr(>|z|)
obesityTRUE         -0.0649034  0.9371580  0.0278814 -2.328  0.01992 *
ageHosp              0.0206863  1.0209017  0.0004869 42.489  < 2e-16 ***
cardioTRUE          -0.0868785  0.9167885  0.0150834 -5.760 8.42e-09 ***
pneumopathyTRUE      0.0641097  1.0662094  0.0231726  2.767  0.00566 **
immunoTRUE           0.1424290  1.1530712  0.0260728  5.463 4.69e-08 ***
raceBlack            0.0977117  1.1026449  0.0766013  1.276  0.20210
raceIndigenous      -0.0309233  0.9695500  0.2244192 -0.138  0.89040
raceMixed            0.0345585  1.0351626  0.0705723  0.490  0.62435
raceWhite            0.0031582  1.0031632  0.0699865  0.045  0.96401
```

```
influenzaVacineUnknown  0.0664629  1.0687213  0.0162767  4.083 4.44e-05 ***
influenzaVacineYes     -0.0446423  0.9563395  0.0237066 -1.883  0.05968 .
covidVacineUnknown     -0.1255183  0.8820397  0.0522272 -2.403  0.01625 *
covidVacineYes         -0.0903383  0.9136221  0.0187867 -4.809 1.52e-06 ***
renalTRUE               0.1207058  1.1282930  0.0232817  5.185 2.16e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                       exp(coef) exp(-coef) lower .95 upper .95
obesityTRUE               0.9372     1.0671    0.8873    0.9898
ageHosp                   1.0209     0.9795    1.0199    1.0219
cardioTRUE                0.9168     1.0908    0.8901    0.9443
pneumopathyTRUE           1.0662     0.9379    1.0189    1.1158
immunoTRUE                1.1531     0.8672    1.0956    1.2135
raceBlack                 1.1026     0.9069    0.9489    1.2813
raceIndigenous            0.9695     1.0314    0.6245    1.5052
raceMixed                 1.0352     0.9660    0.9014    1.1887
raceWhite                 1.0032     0.9968    0.8746    1.1507
influenzaVacineUnknown    1.0687     0.9357    1.0352    1.1034
influenzaVacineYes        0.9563     1.0457    0.9129    1.0018
covidVacineUnknown        0.8820     1.1337    0.7962    0.9771
covidVacineYes            0.9136     1.0945    0.8806    0.9479
renalTRUE                 1.1283     0.8863    1.0780    1.1810

Concordance= 0.594  (se = 0.002 )
Likelihood ratio test= 2190  on 14 df,   p=<2e-16
Wald test            = 1917  on 14 df,   p=<2e-16
Score (logrank) test = 1931  on 14 df,   p=<2e-16

> cox.zph(coxCovid)
                 chisq df       p
obesity          0.404  1 0.52485
ageHosp         11.919  1 0.00056
cardio          12.445  1 0.00042
pneumopathy      0.791  1 0.37372
immuno           0.220  1 0.63905
race            80.061  4 < 2e-16
influenzaVacine 17.275  2 0.00018
covidVacine     22.668  2 1.2e-05
renal            1.389  1 0.23856
GLOBAL         117.813 14 < 2e-16
> summary(coxCovid2)
Call:
coxph(formula = covidSurv ~ obesity + ageHosp + cardio + pneumopathy +
    immuno + covidVacine + renal, data = covid, method = "breslow")

  n= 52062, number of events= 18967

                         coef  exp(coef)   se(coef)      z Pr(>|z|)
obesityTRUE        -0.0661054  0.9360322  0.0278584 -2.373  0.01765 *
ageHosp             0.0204395  1.0206498  0.0004828 42.333  < 2e-16 ***
cardioTRUE         -0.0863952  0.9172317  0.0150775 -5.730 1.00e-08 ***
pneumopathyTRUE     0.0669923  1.0692872  0.0231600  2.893  0.00382 **
immunoTRUE          0.1474518  1.1588775  0.0260506  5.660 1.51e-08 ***
covidVacineUnknown -0.0986695  0.9060421  0.0519098 -1.901  0.05733 .
covidVacineYes     -0.0975382  0.9070677  0.0185523 -5.257 1.46e-07 ***
```

```
renalTRUE           0.1245595  1.1326494  0.0232675  5.353 8.63e-08 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

                   exp(coef) exp(-coef) lower .95 upper .95
obesityTRUE           0.9360     1.0683    0.8863    0.9886
ageHosp               1.0206     0.9798    1.0197    1.0216
cardioTRUE            0.9172     1.0902    0.8905    0.9447
pneumopathyTRUE       1.0693     0.9352    1.0218    1.1189
immunoTRUE            1.1589     0.8629    1.1012    1.2196
covidVacineUnknown    0.9060     1.1037    0.8184    1.0031
covidVacineYes        0.9071     1.1025    0.8747    0.9407
renalTRUE             1.1326     0.8829    1.0822    1.1855

Concordance= 0.592  (se = 0.002 )
Likelihood ratio test= 2146  on 8 df,   p=<2e-16
Wald test            = 1874  on 8 df,   p=<2e-16
Score (logrank) test = 1888  on 8 df,   p=<2e-16

> cox.zph(coxCovid2)
             chisq df       p
obesity      0.424  1 0.51502
ageHosp     11.317  1 0.00077
cardio      12.335  1 0.00044
pneumopathy  0.988  1 0.32022
immuno       0.196  1 0.65816
covidVacine 22.748  2 1.1e-05
renal        1.351  1 0.24515
GLOBAL      39.301  8 4.3e-06

> summary(coxFlu)
Call:
coxph(formula = fluSurv ~ obesity + ageHosp + cardio + pneumopathy +
    immuno + race + influenzaVacine + covidVacine + renal, data = flu,
    method = "breslow")

  n= 24439, number of events= 4720

                            coef  exp(coef)   se(coef)       z Pr(>|z|)
obesityTRUE            0.3249474  1.3839579  0.0474375   6.850 7.38e-12 ***
ageHosp                0.0171763  1.0173247  0.0006432  26.706  < 2e-16 ***
cardioTRUE             0.0644668  1.0665902  0.0342714   1.881  0.05996 .
pneumopathyTRUE       -0.0475222  0.9535893  0.0376689  -1.262  0.20710
immunoTRUE            -0.0376525  0.9630475  0.0577339  -0.652  0.51429
raceBlack              0.2835991  1.3279005  0.1773234   1.599  0.10975
raceIndigenous         0.6653454  1.9451623  0.2735733   2.432  0.01501 *
raceMixed              0.3826343  1.4661417  0.1667531   2.295  0.02176 *
raceWhite              0.2760225  1.3178775  0.1655284   1.668  0.09541 .
influenzaVacineUnknown  0.0310276  1.0315139  0.0343962   0.902  0.36702
influenzaVacineYes     -0.4601093  0.6312146  0.0441490 -10.422  < 2e-16 ***
covidVacineUnknown     -0.3884446  0.6781108  0.1446427  -2.686  0.00724 **
covidVacineYes         -0.2990980  0.7414867  0.0468176  -6.389 1.67e-10 ***
renalTRUE              0.1013462  1.1066597  0.0590938   1.715  0.08634 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
                       exp(coef) exp(-coef) lower .95 upper .95
obesityTRUE               1.3840     0.7226    1.2611    1.5188
ageHosp                   1.0173     0.9830    1.0160    1.0186
cardioTRUE                1.0666     0.9376    0.9973    1.1407
pneumopathyTRUE           0.9536     1.0487    0.8857    1.0267
immunoTRUE                0.9630     1.0384    0.8600    1.0784
raceBlack                 1.3279     0.7531    0.9381    1.8798
raceIndigenous            1.9452     0.5141    1.1379    3.3252
raceMixed                 1.4661     0.6821    1.0574    2.0329
raceWhite                 1.3179     0.7588    0.9527    1.8229
influenzaVacineUnknown    1.0315     0.9694    0.9643    1.1035
influenzaVacineYes        0.6312     1.5842    0.5789    0.6883
covidVacineUnknown        0.6781     1.4747    0.5107    0.9004
covidVacineYes            0.7415     1.3486    0.6765    0.8127
renalTRUE                 1.1067     0.9036    0.9856    1.2426

Concordance= 0.64  (se = 0.004 )
Likelihood ratio test= 1122  on 14 df,   p=<2e-16
Wald test            = 1029  on 14 df,   p=<2e-16
Score (logrank) test = 1067  on 14 df,   p=<2e-16

> cox.zph(coxFlu)
                   chisq df       p
obesity          7.05835  1 0.00789
ageHosp          9.38464  1 0.00219
cardio           5.90519  1 0.01510
pneumopathy     11.52159  1 0.00069
immuno           5.64659  1 0.01749
race            16.55174  4 0.00236
influenzaVacine 30.85752  2   2e-07
covidVacine      9.98409  2 0.00679
renal            0.00191  1 0.96518
GLOBAL          87.71659 14   1e-12
> summary(coxFlu2)
Call:
coxph(formula = fluSurv ~ obesity + ageHosp + race + influenzaVacine,
    data = flu)

  n= 24439, number of events= 4720

                            coef exp(coef)  se(coef)       z Pr(>|z|)
obesityTRUE             0.333392  1.395695  0.047254   7.055 1.72e-12 ***
ageHosp                 0.016656  1.016796  0.000583  28.570  < 2e-16 ***
raceBlack               0.272898  1.313767  0.177277   1.539   0.1237
raceIndigenous          0.673821  1.961720  0.273568   2.463   0.0138 *
raceMixed               0.361047  1.434831  0.166693   2.166   0.0303 *
raceWhite               0.273109  1.314044  0.165500   1.650   0.0989 .
influenzaVacineUnknown -0.017424  0.982727  0.033587  -0.519   0.6039
influenzaVacineYes     -0.472400  0.623504  0.044048 -10.725  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

                       exp(coef) exp(-coef) lower .95 upper .95
obesityTRUE               1.3957     0.7165    1.2722    1.5311
ageHosp                   1.0168     0.9835    1.0156    1.0180
raceBlack                 1.3138     0.7612    0.9282    1.8596
```

```
raceIndigenous            1.9617     0.5098    1.1476    3.3535
raceMixed                 1.4348     0.6969    1.0349    1.9893
raceWhite                 1.3140     0.7610    0.9500    1.8175
influenzaVacineUnknown    0.9827     1.0176    0.9201    1.0496
influenzaVacineYes        0.6235     1.6038    0.5719    0.6797

Concordance= 0.634  (se = 0.004 )
Likelihood ratio test= 1087  on 8 df,   p=<2e-16
Wald test            = 991.1  on 8 df,   p=<2e-16
Score (logrank) test = 1026  on 8 df,   p=<2e-16

> cox.zph(coxFlu2)
                chisq df       p
obesity          7.15  1  0.0075
ageHosp          9.98  1  0.0016
race            15.93  4  0.0031
influenzaVacine 29.93  2 3.2e-07
GLOBAL          61.20  8 2.7e-10


> survdiff(combinedSurv ~ source, data = combined, rho = 0)
Call:
survdiff(formula = combinedSurv ~ source, data = combined, rho = 0)

                 N Observed Expected (O-E)^2/E (O-E)^2/V
source=Covid 52062    18967    16380       409      1367
source=Flu   24439     4720     7307       916      1367

 Chisq= 1367  on 1 degrees of freedom, p= <2e-16
> survdiff(combinedSurv ~ source, data = combined, rho = 1)
Call:
survdiff(formula = combinedSurv ~ source, data = combined, rho = 1)

                 N Observed Expected (O-E)^2/E (O-E)^2/V
source=Covid 52062    14466    12524       301      1229
source=Flu   24439     3616     5558       678      1229

 Chisq= 1229  on 1 degrees of freedom, p= <2e-16


> summary(combinedCox)
Call:
coxph(formula = combinedSurv ~ source, data = combined, method = "breslow")

  n= 76501, number of events= 23687

              coef exp(coef) se(coef)      z Pr(>|z|)
sourceFlu -0.58488   0.55717  0.01628 -35.93   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

          exp(coef) exp(-coef) lower .95 upper .95
sourceFlu    0.5572      1.795    0.5397    0.5752

Concordance= 0.555  (se = 0.002 )
Likelihood ratio test= 1440  on 1 df,   p=<2e-16
```

```
Wald test            = 1291  on 1 df,   p=<2e-16
Score (logrank) test = 1328  on 1 df,   p=<2e-16

> cox.zph(combinedCox)
       chisq df     p
source  4.95  1 0.026
GLOBAL  4.95  1 0.026
```

3.2. **Generative AI Usage.** I primarily used AI to aid me with debugging my code. All of the code was originally written by yours truly, and then if I ran into issues which I could not solve on my own, I used generative AI to edit the code that I had already written. I also used it as a secondary search engine but none of the written part of my text was written by AI.